BLOOM PUBLIC SCHOOL

Vasant Kunj, New Delhi

Sample Question Paper

Mid-Term Examination(2026-27)

Mathematics

Class –VI

Date: 28/09/2026

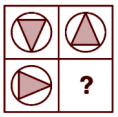
Time Allowed:2.5 hours **Max Marks- 60**

General Instructions:

1. This Question Paper has 5 Sections A, B, C, D and E.
2. Section A has 10 MCQs and 6 fill ups carrying 1 mark each(from Q1 - 16).
3. Section B has 4 questions carrying 02 marks each(from Q17 - 20).
4. Section C has 3 questions carrying 03 marks each(from Q21 - 23).
5. Section D has 3 questions carrying 05 marks each(from Q24 - 26).
6. Section E has 3 case-based integrated units of assessment (04 marks each) with sub-parts of the values of 1, 1 and 2 marks each respectively(from Q27 - 29).
7. All Questions are compulsory. However, an internal choice of 1 Question of 5 marks, 1 Question of 3 marks and 1 Question of 2 marks has been provided. An internal choice has been provided in the 2 marks questions of Section E. You have to attempt only one of the alternatives in all such questions.

SECTION A

Q1. Observe the following figure and choose the correct option. **1**



(a)  (b)  (c)  (d) 

Q2. If two lines intersect, how many points do they have in common? **1**

(a) None (b)One (c) Two (d)Infinite

Q3. Which of these numbers is **not** a palindrome? **1**

(a)121 (b)454 (c)678 (d)2002

Q4. A pictograph represents data in the form of: **1**

(a)vertical bar (b) horizontal bar (c)pictures (d)tally marks

Q5. Which of the following numbers is divisible by 5 and 10 **both**? **1**

(a)120 (b)135 (c)245 (d)579

Q6. Which of the following is the best example of an angle in real life? **1**

(a)closed book edge (b)pencil tip (c)open door (d)spoon curve

Q7. In a pattern of dots arranged like this: ••, •••, ••••, how many dots will be in the next group? **1**

(a) • (b)•••• (c)•• (d)•••••

Q8. Which tool is used to measure angles? **1**

(a)Ruler (b)Compass (c)Protractor (d)Divider

Question No. - 9 and 10 consist of two statements- Assertion (A) and Reason(R). Answer the questions by selecting appropriate options.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is False but R is true.

Q9. **Assertion(A):** 17 is a composite number. **1**
Reason(R): Numbers having more than two factors are called composite numbers.

Q10. **Assertion(A):** The pattern 1, 4, 9, 16, 25 represents prime numbers. **1**
Reason(R): Each number in the pattern above is the square of a natural number.

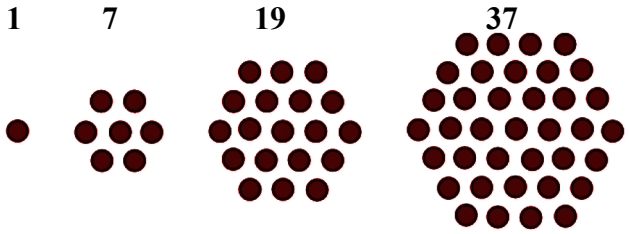
Fill in the blanks from Q.11-16

Q11. There are _____ 1-digit numbers. **1**

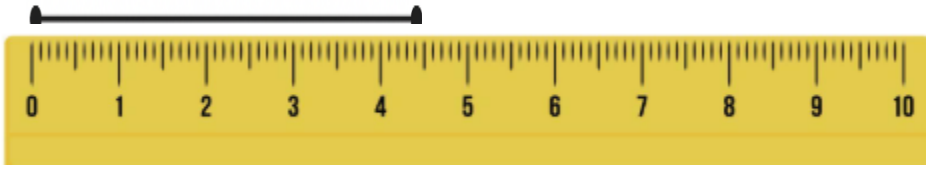
Q12. The number(s) which is/are supercell(s) in the table below is/are _____. **1**

200	199	345	202	300	180	271	198

Q13. In the following sequence, the numbers are called _____. **1**



Q14. The measure of the given line segment is _____ cm. **1**

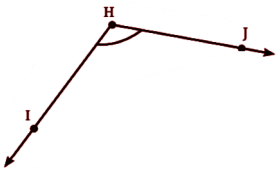


Q15. An example of twin primes is the pair of numbers: ___, ___. **1**

Q16. Identify the pattern for the following sequence and find the next letter. **1**
A, C, E, G, ___.

SECTION B

Q17. Use a protractor to draw an angle whose degree measure is the same as the angle given below: **2**

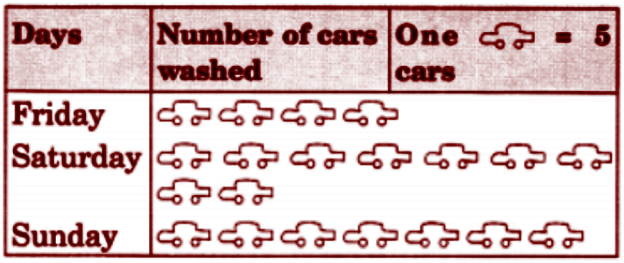


Q18.(I) Find the palindromic number by using the following 2-digit number: 10 **2**

OR

(II) Check if the Collatz Conjecture holds for the starting number 10.

Q19. Mr. Rajan made a pictograph given below to show the number of cars washed at a car washing station during three days of a week. **2**



From the pictograph, find that:

(i) How many cars were washed on Friday.

(ii)How many more cars were washed on Saturday than on Friday?

Q20. Draw two different factor trees for 144. **2**

SECTION C

Q21. The first five terms of a number pattern are: 5, 10, 20, 40, 80 **3**

(i)What is the rule used to form this pattern?

(ii)Write the next two terms of the pattern.

(iii)If the first term is 6, find the 4th term applying the same rule.

Q22. The result of a Mathematics test is as follows: **3**

80, 90, 70, 80, 80, 60, 80, 70, 90, 65, 100, 60, 70, 60, 70, 85, 65, 70, 70, 85, 90, 60, 65, 80, 60. Make a frequency table for the above data and write how many students appeared in the test?

Q23.(I) Use prime factorisation to verify whether 35 and 84 are co-prime. **3**

OR

(II)Find the prime factorisation of the number without multiplying: 56 × 25.

SECTION D

Q24 (i) Write the smallest 3-digit palindrome and the largest 4-digit palindrome. **5**

(ii)Write one 5-digit number and two 3-digit numbers such that their sum is 24,530.

(iii)Express 432 as the product of prime factors.

Q25. The following data represents the sale of refrigerator sets in a showroom in the first 6 months of the year. **5**

Months	Jan	Feb	March	April	May	June
No. of Refrigerators Sold	20	25	15	40	35	30

Draw the bar graph for the data given and find out the months in which the sale was minimum and maximum.

Q26. Answer the following: **5**

(I) (i) Find the common factors of 72 and 150.

(ii) Is 150 divisible by 25? Check by prime factorisation.

(iii) What is the smallest number whose prime factorization has three different prime numbers?

OR

(II) (i) Identify two numbers between 21 and 49 that are co-prime.

(ii) The prime factorisation of a number has two 5s and three 2s. What is the number?

(iii)Check with the divisibility rule whether 1368 is divisible by 8. Show your calculations.

SECTION E

Case Study-I

Q27. Kabir organized a fun classroom challenge during the mathematics exhibition where students had to guess the angles formed between the hour and minute hands of a clock at different times. He asked everyone to calculate the exact angle at specific hours without using a protractor.

(i) What is the angle formed at 2pm? **1**

(ii) Find the angle formed at 5pm. **1**



(iii) (A)Draw, name and classify the angle formed in (i). **2**

OR

(B)Find the reflex of the angle formed in (ii).

Case Study-II



Q28. An archaeologist finds a tablet detailing an ancient encryption method. The process, which they name the "Kaprekar Scramble", works on 4-digit keys. The tablet shows the key 3087 was scrambled.

(i) To begin the scramble of key 3087, what is the largest number that can be formed from its digits? **1**

(ii) What is the smallest number that can be formed from the digits of 3087? **1**

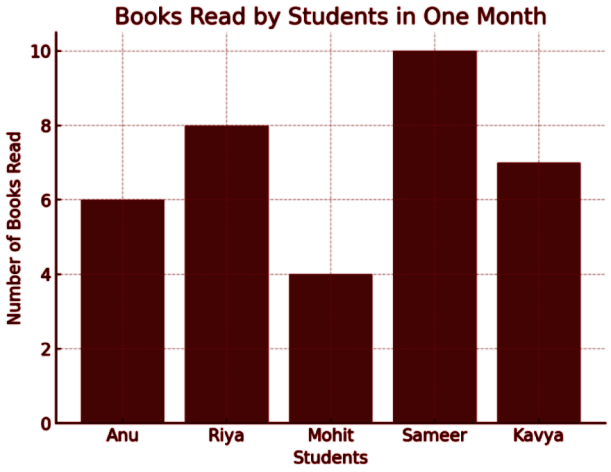
(iii) (A) What is the stable "magic number" that the 4-digit Kaprekar Scramble always converges to? Calculate the result of the first round of the Kaprekar Scramble on the key 3087 by performing the correct subtraction. **2**

OR

(B) Calculate the Kaprekar's constant on your birth year by clearly showing each step.

Case Study-III

Q29. Observe the graph below and answer the following questions:



(i) Find the total number of books read by all students. **1**

(ii) How many students read more than 6 books? **1**

(iii)(A)How many books did Riya and Kavya read together? **2**

OR

(B) What is the difference between the books read by Sameer and Mohit?